Two-Sided Certificates for Real Littlewood Polynomials Join Defects, Sparse Rounding, and Physical Spectral Bounds

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Abstract

The assigned source claims real Littlewood polynomials with modulus arbitrarily close to the square-root length at every sufficiently large integer length. We read its complete balanced-function, oscillatory-wave, rounding, and completion argument while retaining the companion packing and real-discrepancy lemmas as imported inputs. We quantify the exterior Fourier-tail cost of imperfect leading-phase derivative joins, with exact continuity still required, and combine this with a finite Parseval-defect certificate preserving a positive lower modulus. A refinement of the source's dyadic rounding removes its additive truncation constant and records a sparse flip budget; this is an existence statement, not a new efficient discrepancy algorithm. Exact autocorrelation certificates then give physical bilateral-convolution and orbit-Gram bounds, including finite signed spectral products. Positive tower coefficients, Koopman realization, moving-window convergence, and cyclicity on the intended probability space remain distinct obligations. Contribution assessment: C3/5, a source-specific quantitative join-to-rounding lower-floor certificate with proved physical consequences, not a new global ultraflatness proof or a solution of Banach's spectral problem.

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1 Normalization, status, and proof boundaries

We use $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ with Haar probability measure dt and $\mathrm{e}(t) = \exp(2\pi i t)$. For a function f on $\mathbb{T}$, $\widehat{f}(k) = \int_{\mathbb{T}} f(t)\mathrm{e}(-kt)\,dt$. A real Littlewood polynomial of length N is

$$P(z) = \sum_{j=0}^{N-1} \varepsilon_j z^j, \qquad \varepsilon_j \in \{-1, 1\}; \qquad p(t) = N^{-1/2} P(\mathrm{e}(t)).$$

Thus $\int |p|^2 = 1$. Its degree is $N-1$, not N. Two-sided ϵ-ultraflatness means $1 - \epsilon \le |p(t)| \le 1 + \epsilon$ for *all* $t \in \mathbb{T}$, including 0 and $1/2$. Signs may be chosen independently at each length.

The assigned October 5, 2026 manuscript [1] claims this property for every fixed $\epsilon > 0$ and all sufficiently large N. That is a stronger statement than the published constant-factor flatness theorem of Balister et al. [3], whose exact Theorem 1.1 gives fixed positive lower and upper multiples of $\sqrt{n}$ for every degree $n \ge 2$. The present report does not turn source provenance, a complete literal read, or small finite checks into independent verification of a new global existence theorem.

1.1 Current primary-literature checks

Erdős-type lower obstructions are consistent with an error tending to zero. Erdélyi's August 2026 preprint [4], Theorems 2.1 and 2.2, states, in our length normalization,

$$\max |P|^2 \ge N + (N-1)^{1/3}/38, \tag{1}$$

$$\max\{\max |P|^2 - N,\ N - \min |P|^2\} \ge \sqrt{(N-1)/2}. \tag{2}$$

Consequently a symmetric error certificate must obey $(2\epsilon + \epsilon^2)N \ge \sqrt{(N-1)/2}$ if that cited result is invoked. This is a literature consequence, not a local contribution or an independently re-proved external theorem. The exact statements were inspected; their complete proof dependencies were not audited.

Erdélyi's separate fixed-prefix theorem [5] states that all partial sums of one fixed infinite unimodular coefficient sequence cannot form an ultraflat sequence. Its exact statement was inspected in the author's text. It does not forbid independent choices at each length, and it does not justify treating the source's choices as compatible prefixes of a single infinite sign sequence. These distinctions persist when building products or proposing a dynamical realization.

1.2 Literal source scope and imported inputs

All seven assigned TeX files, the bibliography, and README were read literally in bounded displays: nine public text files, 1,333 lines. The scientific architecture is as follows.

1. `oscillation.tex`: a uniform quadratic formula, compactly supported stationary phase, and endpoint-amplitude variation bounds.

2. `rounding.tex`: dyadic defect-sensitive rounding with prescribed complex phases; real inputs give real signs. Its real matrix discrepancy lemma is imported from the companion paper [2].

3. `balanced.tex`: a bounded real torus polynomial, frequency-box spreading, and a common vector giving weights $w_a = |a \cdot v|$.

4. `waves.tex`: signed interval packing, coherent stationary contributions, large endpoint curvature, gap filling, and exterior-tail cancellation. Signed interval packing is the other imported companion lemma.

5. `completion.tex`: real Fourier coefficients, Parseval control of the sign deficit, projection, and uniform two-sided completion.

The introduction and main file specify the all-length quantifiers. We read the two imported lemmas as stated in this manuscript, but did not independently read or certify their proofs in the companion source. They are not silently counted among our local proofs. No other author's report is a premise here.

2 The source construction, with its costs retained

The source first constructs $F : \mathbb{T}^m \to \mathbb{R}$ with finite symmetric nonzero frequency support $\mathcal{A}$, at least two nonparallel sign pairs, and a vector $v \in \mathbb{R}^m$ such that

$$\|F\|_\infty \le 1+\delta, \quad 1 - C\delta \le \sum_a w_a < 1, \quad 1 \le \frac{|c_a|}{\sqrt{w_a}} \le 1 + C\delta \le 2, \quad w_a = |a \cdot v| > 0. \tag{3}$$

These are weights induced by one vector, not independently assigned numbers. Starting from $p_j = p_{j-1} + \frac{1}{2}(1 - p_{j-1}^2)\cos(2\pi y_j)$, the source proves $|p_j| \le 1$ and $M_j - M_{j-1} \ge (1 - M_{j-1})^2/8$ for $M_j = \int p_j^2$. Thus $M_j \uparrow 1$. This elementary recursion is retained, not credited as a new result.

Each original frequency s is spread into a box of size $n_s \sim T^D \prod_j |s \cdot W_j|$. The vectors W_j are chosen so $\prod_j |s \cdot W_j| \sim \tau |b_s|^2/|s \cdot \gamma|$. Complex rounding makes every coefficient over that box have the exact magnitude $|b_s|/\sqrt{n_s}$. Conjugating the negative-frequency factors preserves reality. The source's explicit bounds are

$$\frac{(1-\delta)^3}{(1+\delta)^3} \le \sum_a w_a \le \frac{1}{1+\delta}, \qquad 1 + \delta \le |c_a|^2/w_a \le \frac{(1+\delta)^3}{(1-\delta)^2}.$$

All dimensions and spreading data are fixed before the final length N.

Signed interval packing supplies disjoint arcs centered at $t^*_{a,h} \equiv -a \cdot \theta_h$. With $\lambda_a = a \cdot v$, the source parametrizes the inverse phase derivative by

$$T_{a,h}(x) = t^*_{a,h} - \lambda_a Q_h'(x), \qquad Q_h'' = \chi_h^2.$$

The leading phase obeys $\psi' = X_{a,h}$ and $\psi'' = -1/(\lambda_a \chi_h^2)$. The wave

$$\frac{c_a}{\sqrt{w_a}} \mathrm{e}\big(\operatorname{sgn}(\lambda_a)/8 + N\psi(t)\big)$$

has constant modulus near one. Its stationary contribution at $x = k/N$ is $c_a G_h(x)\chi_h(x)\mathrm{e}(-kt^*_{a,h} + N\lambda_a Q_h(x))$. Summing over a gives one value $G_h\chi_h F(k\theta_h + NvQ_h)$; it does not cost $\sum_a |c_a|$. This coherence depends on integer Fourier indices and the common vector v.

At the ends, small χ_h increases curvature without decreasing the wave's modulus. Gaps have total length $O(\delta)$, bounded amplitude variation, large outer slopes, and separated middle derivative intervals. These produce a scaled Fourier cost $O(\sqrt{\delta}) + o(1)$. Bounded affine phase corrections, which depend on N, match endpoint values; leading phase derivatives match as well. The source obtains a continuous, conjugate-symmetric B_N with

$$1 \le |B_N| \le 1 + C\delta, \quad \sqrt{N}|\widehat{B}_N(k)| \le 1 + C\sqrt{\delta}\ (0 \le k < N), \quad \sum_{k \notin [0,N-1]} |\widehat{B}_N(k)| \le K_\delta/N. \quad (4)$$

Full derivatives of B_N need not match at joins. Leading derivative matching cancels the first integration-by-parts boundary terms. That specific mechanism is the subject of the next quantitative extension.

3 A stable exterior-tail bound for imperfect joins

Let $N \ge 1$ be an integer. Partition the circle into K nondegenerate closed intervals with disjoint interiors. On piece I_j write

$$B(t) = f_j(t)\mathrm{e}(N\phi_j(t)), \quad (5)$$

where $f_j \in C^2(I_j;\mathbb{C})$ and $\phi_j \in C^3(I_j;\mathbb{R})$, with one-sided derivatives at endpoints. Suppose B itself is continuous and periodic. Assume the explicit uniform bounds

$$\sigma \le \phi_j' \le 1 - \sigma, \quad \max(\|\phi_j''\|_\infty, \|\phi_j'''\|_\infty) \le \Lambda, \quad \max_{0 \le a \le 2} \|f_j^{(a)}\|_\infty \le M, \quad (6)$$

where $0 < \sigma \le 1/2$, $M > 0$, $\Lambda \ge 0$. The amplitudes and phases may depend on N if these bounds are retained; no hidden uniformity in shrinking gaps is assumed. At each join, including the periodic join, let ν_-, ν_+ be the left and right leading phase derivatives. Put

$$\mathcal{J} = \sum_{\text{joins } t_j} |B(t_j)|\,|\nu_+(t_j) - \nu_-(t_j)|, \qquad c = \sigma/2, \quad (7)$$

and define the fully specified constant

$$C_2 = \frac{M}{4\pi^2}\left((2K+1)c^{-2} + (2K+4)\Lambda c^{-3} + 3\Lambda^2 c^{-4}\right). \quad (8)$$

Theorem 3.1 (Join-defect exterior certificate)**.** *Under these assumptions, for every exterior integer $k < 0$ or $k \ge N$,*

$$|\widehat{B}(k)| \le \left(\frac{N\mathcal{J}}{2\pi c^2} + C_2\right)(N + |k|)^{-2}. \quad (9)$$

In particular its exterior absolute Fourier tail satisfies

$$\tau := \sum_{k<0 \text{ or } k \ge N} |\widehat{B}(k)| \le \frac{\mathcal{J}}{\pi c^2} + \frac{2C_2}{N}. \quad (10)$$

The Fourier series converges absolutely and represents B everywhere.

Proof. Write $D = N + |k|$ and $V_j = N\phi_j' - k$. For $k < 0$, $|V_j| \ge \sigma D$; for $k \ge N$, $|V_j| \ge k - (1-\sigma)N \ge cD$. Also $|V_j'|, |V_j''| \le N\Lambda$. One integration by parts on each interval has boundary value

$$\frac{B(t)\mathrm{e}(-kt)}{2\pi i V_j(t)}.$$

At a join, continuity gives the same numerator on both sides. Its net absolute size is at most

$$\frac{|B(t_j)|}{2\pi}\left|\frac{1}{N\nu_- - k} - \frac{1}{N\nu_+ - k}\right| \le \frac{N|B(t_j)|\,|\nu_+ - \nu_-|}{2\pi c^2 D^2}.$$

Periodicity and integer k give the same calculation at the circle's identified endpoints. Summing yields the first term in (9).

Integrate the remainder once more. With $A = (f_j/V_j)'/V_j = f_j'/V_j^2 - f_j V_j'/V_j^3$, its derivative is

$$A' = \frac{f_j''}{V_j^2} - \frac{3f_j'V_j'}{V_j^3} - \frac{f_jV_j''}{V_j^3} + \frac{3f_j(V_j')^2}{V_j^4}.$$

Since $N \le D$, we have

$$|A| \le M(c^{-2} + \Lambda c^{-3})D^{-2}, \quad |A'| \le M(c^{-2} + 4\Lambda c^{-3} + 3\Lambda^2 c^{-4})D^{-2}.$$

There are $2K$ second boundary terms; the total interval length in the remaining integrals is one. Including the factor $(2\pi)^{-2}$ gives $C_2 D^{-2}$. Finally, $\sum_{k<0 \text{ or } k\ge N}(N + |k|)^{-2} \le 2/N$. This follows by comparison with the integral for the negative tail and the integral plus first term for the positive tail. It proves (10). The finitely many retained coefficients and summable exterior coefficients give an absolutely convergent Fourier series; continuity and uniqueness of Fourier coefficients identify its sum with B. □

Meaning and cost of the extension. The source has $\mathcal{J} = 0$. A sequence of imperfect joins with $\mathcal{J}_N \to 0$ is allowed provided $C_{2,N}/N \to 0$ and the displayed constants are used. A fixed nonzero mismatch generally leaves a fixed error budget. Large derivatives from small gaps increase C_2; the theorem does not assert uniformity after changing all auxiliary data. It controls the exterior tail only. The separate coherent stationary coefficient bound in (4) must still be supplied.

Proposition 3.2 (Why a value jump is different)**.** *A piecewise continuous function with a genuine nonzero one-sided jump cannot have an absolutely summable Fourier sequence. Thus continuity in Theorem 3.1 cannot simply be replaced by a small nonzero value-jump tolerance while retaining its conclusion.*

Proof. An absolutely summable sequence gives a continuous periodic Fourier sum. If those are the given function's Fourier coefficients, uniqueness in L^1 makes the two functions equal almost everywhere. On each interval of continuity they must then agree everywhere. Their one-sided limits at a join would coincide, contradicting the genuine jump. □

4 Rounding without a truncation constant

We use exactly the following imported real discrepancy statement, which is quoted in the source from its companion [2]: there is an absolute C_D such that for $1 \le s \le R$ and every real $A \in [-1,1]^{R\times s}$ some $\xi \in \{-1,1\}^s$ satisfies

$$\|A\xi\|_\infty \le C_D\sqrt{s\log(2R/s)}. \tag{11}$$

We do not infer an efficient implementation or a numerical value of C_D merely from this existence statement. Our refinement is a short limit argument within the source's particular mass-preserving rounding.

Theorem 4.1 (Zero-offset rounding and a flip budget)**.** *Let $n \ge 1$, $|Z_j| \le 1$, $Z_j = d_j|Z_j|$ with $|d_j| = 1$, taking $d_j = 1$ when $Z_j = 0$. Define*

$$\mu = \frac{1}{2}\sum_{j=0}^{n-1}(1 - |Z_j|), \qquad C_R = \frac{2\sqrt{2}(1+\sqrt{2})C_D}{1 - \pi\exp(1)/20}.$$

There are $\eta_j \in \{d_j, -d_j\}$ with at most $\lfloor \mu \rfloor$ flips from d_j and

$$\sup_{t\in\mathbb{T}}\left|\sum_j (\eta_j - Z_j)\mathrm{e}(jt)\right| \le \min\left\{2\mu,\ C_R\sqrt{\mu\log(80n/\mu)}\right\}. \tag{12}$$

At $\mu = 0$ the second expression is interpreted as zero. Real inputs give real signs. For $\mu < 1$ the unflipped vector itself satisfies the bound. Consecutive shifted frequencies have the same estimate.

Proof. The case $\mu = 0$ is immediate. Put $p_j = (1 - |Z_j|)/2$, so $0 \le p_j \le 1/2$ and $\sum p_j = \mu$. At the grid $t_\ell = \ell/(20n)$, form $40n$ real rows from real and imaginary parts of $d_j\mathrm{e}(jt_\ell)$. For an arbitrary large integer J, round each p_j down to a multiple of 2^{-J}. This changes every row sum by at most $n2^{-J}$ and does not increase total mass.

For $h = J, J-1, \dots, 1$, use (11) on coordinates whose current numerator on the 2^{-h} grid is odd. Reverse the entire coloring if necessary to make its sum nonpositive, then add $2^{-h}\xi$ on those coordinates. Both choices stay in $[0,1]$, the next coarser grid is reached, and mass remains at most μ. If s_h coordinates are active, each has mass at least 2^{-h}, so $s_h \le \min(n, 2^h\mu)$. Monotonicity of $s\log(80n/s)$ on $(0, n]$ gives

$$s_h\log(80n/s_h) \le 2^h\mu\log(80n/\mu).$$

For $2^h\mu > n$ the left side is at most $n\log 80$ and the same conclusion holds. The final vector $p' \in \{0,1\}^n$ therefore has mass at most μ and matrix discrepancy at most

$$n2^{-J} + C_D(1+\sqrt{2})\sqrt{\mu\log(80n/\mu)},$$

using $\sum_{h\ge1} 2^{-h/2} = 1 + \sqrt{2}$.

There are only finitely many possible final binary vectors. Along an infinite subsequence $J \to \infty$ one of them is constant. Passing to the limit removes $n2^{-J}$ without changing the mass bound. Set $\eta_j = d_j(1 - 2p'_j)$. Each real and imaginary grid error is at most twice the matrix bound, giving the factor $2\sqrt{2}$.

For the error polynomial Q of degree at most $n-1$, let $S_Q = \max_{|z|=1}|Q(z)|$. The maximum principle, also applied to the reversed polynomial, gives $|Q(z)| \le S_Q\max(1, |z|^{n-1})$. Cauchy's estimate on radius-$1/n$ disks centered on the circle implies $|(d/dt)Q(\mathrm{e}(t))| \le 2\pi\exp(1)nS_Q$. Every point is within $1/(40n)$ of the grid. Therefore $S_Q \le \max_\ell|Q(\mathrm{e}(t_\ell))| + (\pi\exp(1)/20)S_Q$. This proves the second bound in (12).

The choice $p' = 0$ always has error at most $\sum(1 - |Z_j|) = 2\mu$ and no flips. Choose whichever of these two constructions has the smaller certified bound. Both have at most $\lfloor\mu\rfloor$ flips. If $\mu < 1$, the discrepancy construction also has no flips, so the assertions are consistent. Multiplication by a monomial handles a shifted interval; real phases are signs. □

Increment versus source. The source stops its initial dyadic truncation when $n2^{-J} \le 1$, and states $C(1 + \sqrt{\mu\log(80n/\mu)})$. The finite-output limit removes that additive constant. Its proof already contains the nonincreasing-mass mechanism; we expose its exact flip consequence and retain the elementary 2μ alternative. This improves very small defects and removes a finite $N^{-1/2}$ term after normalization. It does not improve the source's main δ exponent or certify a fast algorithm. The refinement alone would not justify a high contribution score.

5 A finite two-sided completion certificate

Let $B : \mathbb{T} \to \mathbb{C}$ be continuous with absolutely summable Fourier series and conjugate symmetry $B(-t) = \overline{B(t)}$. Its Fourier coefficients are real. Suppose supplied bounds satisfy

$$\beta \le |B(t)| \le \Gamma, \quad \sqrt{N} \max_{0 \le k < N} |\widehat{B}(k)| \le S, \quad S > 0, \qquad \sum_{k \notin [0, N-1]} |\widehat{B}(k)| \le \tau. \tag{13}$$

The numbers β, Γ, τ are nonnegative. Set

$$Y_k = \sqrt{N}\widehat{B}(k)/S, \quad U_Y = N^{-1/2} \sum_{k=0}^{N-1} Y_k \mathrm{e}(kt), \quad \mu = \tfrac{1}{2} \sum_k (1 - |Y_k|). \tag{14}$$

All quantities are finite; there is no implicit asymptotic $o(1)$.

Theorem 5.1 (Wave-to-sign lower-floor budget)**.** *There are real signs with $p = N^{-1/2} \sum \varepsilon_k \mathrm{e}(kt)$ and at most $\lfloor \mu \rfloor$ flips relative to the nearest-sign phases of Y, such that*

$$L := \frac{\beta - \tau}{S} - r_N(\mu) \le |p(t)| \le \frac{\Gamma + \tau}{S} + r_N(\mu) =: U, \tag{15}$$

$$r_N(\mu) = \min\left\{ \frac{2\mu}{\sqrt{N}}, \quad C_R \sqrt{\frac{\mu}{N} \log(80N/\mu)} \right\}. \tag{16}$$

The zero-defect convention is as before. In particular $L > 0$ is a sufficient finite certificate of no zeros anywhere on the unit circle. If $E_0 \le \|B\|_2^2$ is a known energy lower bound, then

$$\frac{\mu}{N} \le \frac{1}{2}\left(1 - \frac{\|B\|_2^2 - \sum_{k \notin [0,N-1]} |\widehat{B}(k)|^2}{S^2}\right) \le \frac{1}{2}\left(1 - \frac{\max(0, E_0 - \tau^2)}{S^2}\right). \tag{17}$$

The rightmost bound is nonnegative under the hypotheses.

Proof. Absolute summability gives $\|U_Y - B/S\|_\infty \le \tau/S$. Theorem 4.1 gives $\|p - U_Y\|_\infty \le r_N(\mu)$; both sides of (15) follow from the triangle and reverse triangle inequalities. For the defect, $|Y_k| \ge Y_k^2$ and Parseval give the exact identity

$$\frac{1}{N} \sum_k Y_k^2 = \frac{\|B\|_2^2 - \sum_{k \notin [0,N-1]} |\widehat{B}(k)|^2}{S^2}.$$

The exterior squared sum is at most τ^2, not merely a quantity linear in τ. The retained energy is nonnegative and at most S^2, because each $|Y_k| \le 1$. These facts prove (17). □

Combining the local improvements. Theorem 3.1 supplies $\tau \le \mathcal{J}/(\pi c^2) + 2C_2/N$ in this certificate, provided conjugate symmetry and the retained coefficient bound are independently maintained. When $\mathcal{J} = 0$, the exterior energy loss is bounded by $4C_2^2/N^2$, while the pointwise projection loss remains $2C_2/N$. The source's stated asymptotic Parseval step does not record this quadratic-versus-linear distinction. If a rounded vector has already been computed, its directly certified uniform error may replace $r_N(\mu)$; existence via discrepancy does not certify any arbitrary output.

For the source's wave profile, take $\beta = 1$, $\Gamma = 1 + C\delta$, $S = 1 + C_1\sqrt{\delta}$, $E_0 = 1$, and $\tau = K_\delta/N$. Then $\mu/N = O(\sqrt{\delta}) + O_\delta(N^{-2})$, and the rounding contribution remains $O(\delta^{1/4}\sqrt{\log(1/\delta)})$ after letting $N \to \infty$ at fixed δ. Our improvements do not give an explicit source length threshold or permit an unproved substitution $\delta = \delta(N)$ through constants K_δ and stationary-phase errors.

Example 5.2 (An exact small-defect floor)**.** Let $P_3(z) = 1 + z - z^2$ and $p_3 = P_3/\sqrt{3}$. Since $|p_3(t)|^2 = 1 - \frac{2}{3}\cos(4\pi t)$, its exact modulus range is $[1/\sqrt{3}, \sqrt{5/3}]$. For $B = (1-\eta)p_3$ with $0 < \eta < 2/3$, use $S = 1$, $\tau = 0$, $Y = (1-\eta)(1, 1, -1)$. Then $\mu = 3\eta/2 < 1$, so the unflipped signs are compulsory under the flip budget. The elementary branch in (16) gives

$$L \geq (1 - 4\eta)/\sqrt{3} > 0 \qquad (0 < \eta < 1/4).$$

The actual error is $\eta\sqrt{5/3}$, smaller than this simple certified budget. This is a concrete finite verification, not a family tending to ultraflatness. The gain concerns arbitrarily small positive defect: the source already handles $\mu = 0$ separately by taking the input itself.

6 Exact real-sign certificates after rounding

For actual signs define the integer aperiodic correlations

$$C_h = \sum_{j=0}^{N-1-h} \varepsilon_j \varepsilon_{j+h} \quad (0 \leq h < N), \qquad C_0 = N. \tag{18}$$

Then the actual normalized squared modulus is

$$q(t) = |p(t)|^2 = 1 + \frac{2}{N} \sum_{h=1}^{N-1} C_h \cos(2\pi h t). \tag{19}$$

This supplies an exact polynomial certificate for both sides, rather than independent upper and lower claims about different rounded vectors.

Proposition 6.1 (Squared-modulus grid budget)**.** *Let $M_g \geq 1$ and suppose certified intervals $[l_j, u_j]$ contain $q(j/M_g)$ for $0 \leq j < M_g$. Put*

$$D_2 = \frac{8\pi^2}{N} \sum_{h=1}^{N-1} h^2 |C_h|, \quad a = \min_j l_j - \frac{D_2}{8M_g^2}, \quad b = \max_j u_j + \frac{D_2}{8M_g^2}.$$

Then $a \leq q(t) \leq b$ on the whole circle. In particular $a > 0$ certifies the modulus floor $\sqrt{a}$. The finite trigonometric sums must be enclosed rigorously, not merely sampled in floating point.

Proof. Differentiating (19) twice gives $\|q''\|_\infty \leq D_2$. At a global minimum or maximum on the circle, $q' = 0$. A grid point lies within distance $1/(2M_g)$ of that extremum. Taylor's integral remainder is at most $D_2/(8M_g^2)$. Applying this separately to the two extrema proves the assertion. The zero derivative belongs to the actual extremum, not to an assumed grid minimizer. □

The second-order grid loss is a classical Taylor consequence specialized to exact integer correlations. It does not mean an arbitrary sampled matrix certifies the continuum. Alternatively put $x = \cos(2\pi t)$ and replace $\cos(2\pi h t)$ by its Chebyshev polynomial $T_h(x)$. Exact factorizations or interval polynomial certificates on $[-1, 1]$ can then avoid a grid altogether.

Example 6.2 (A complete rational lower and upper certificate)**.** For $P_5(z) = 1 + z + z^2 - z^3 + z^4$ the correlations are $(C_1, C_2, C_3, C_4) = (0, 1, 0, 1)$. Thus

$$q(x) = 1 - \frac{12}{5}x^2 + \frac{16}{5}x^4, \qquad q(x) - \frac{11}{20} = \frac{16}{5}\left(x^2 - \frac{3}{8}\right)^2,$$

and

$$\frac{9}{5} - q(x) = (1-x^2)\left(\frac{4}{5} + \frac{16}{5}x^2\right) \geq 0 \qquad (-1 \leq x \leq 1).$$

Both extrema are attained. Consequently the exact squared-modulus range is $[11/20, 9/5]$. This finite example tests the entire circle and includes its real endpoints; it is not evidence for an asymptotic family.

6.1 Correlation energy and a classical two-sided variance bound

Parseval applied to (19) gives

$$\int (q-1)^2 = \frac{2}{N^2} \sum_{h=1}^{N-1} C_h^2. \tag{20}$$

If $a \leq q \leq b$ with $a \leq 1 \leq b$, then $(q-a)(b-q) \geq 0$ and $\int q = 1$ imply

$$\int (q-1)^2 \leq (b-1)(1-a). \tag{21}$$

This is the elementary mean-one Bhatia–Davis variance bound, not a new inequality. For a two-sided ϵ-ultraflat certificate with $0 < \epsilon < 1$, $a = (1-\epsilon)^2$, $b = (1+\epsilon)^2$, the upper bound is $4\epsilon^2 - \epsilon^4$. With the convention $\mathrm{MF}(P) = N^2/(2\sum_{h\geq 1} C_h^2)$, this gives $\mathrm{MF}(P) \geq (4\epsilon^2 - \epsilon^4)^{-1}$ when the denominator is positive. This conditional consequence is included only to preserve normalization, not to duplicate a separate merit-factor report or claim an independently constructed high-merit family.

7 Physical convolution, orbit Grams, and signed products

The confidential originals were freshly read, not replaced by their catalogue descriptions. TR8 [7] displays its Fourier substitution and the space-dependent symbol $K(x, \omega)$. Its broad wording that square integrability permits the displayed pointwise Fubini interchange requires care: L^2 alone does not imply absolute double integrability or pointwise Fourier inversion. We do not use that shortcut. For finite convolution below the Fourier identity is first exact on finitely supported vectors, then extended by boundedness and Plancherel. TR248 [8] supplies the physical Gram convention and quantified implementation-error proof; TR250 [9] confines finite moment inversion to a specified physical space and compatible coefficient data. These are classical Hilbert-space tools with explicit domains.

On $\ell^2(\mathbb{Z})$ use $\langle f, g\rangle = \sum f_j \overline{g_j}$, linear in the first argument, and bilateral shift $(Sf)_j = f_{j-1}$. The unitary Fourier map is $\mathcal{F}f(t) = \sum_j f_j \mathrm{e}(jt)$, first for finite support and then in $L^2(\mathbb{T})$. Let $g_j = \varepsilon_j/\sqrt{N}$ on $0 \leq j < N$ and zero otherwise. Its convolution operator is $C_g c = \sum_j g_j S^j c$; no positive-coefficient restriction is imposed.

Theorem 7.1 (Physical lower floor at every finite window)**.** *Suppose an actual polynomial is certified to satisfy $0 < L \leq |p| \leq U$. Then*

$$L\|c\|_2 \leq \|C_g c\|_2 \leq U\|c\|_2 \qquad (c \in \ell^2(\mathbb{Z})). \tag{22}$$

C_g is boundedly invertible on this bilateral space with inverse norm at most L^{-1}. For every finite index set $I \subset \mathbb{Z}$, the physical synthesis $T_I c = \sum_{j\in I} c_j S^j g$ has Gram bounds

$$L^2 I \preceq T_I^* T_I \preceq U^2 I. \tag{23}$$

If an implementation in the same physical metric has $\|\widetilde{T}_I - T_I\| \leq \zeta < L$, its lower Gram bound is $(L-\zeta)^2$. The ideal bilateral orbit of g is cyclic.

Proof. For finite sequences, direct rearrangement gives $\mathcal{F}(C_g c) = p\,\mathcal{F}c$. The finite kernel makes C_g bounded, so density extends the identity to all of ℓ^2. Plancherel and the pointwise bounds prove (22). Multiplication by $1/p$ is bounded on L^2, hence gives a two-sided bounded inverse; no causal inverse is asserted. Zero-extension from $\mathbb{C}^I$ to ℓ^2 proves (23). The triangle inequality gives $\|\widetilde{T}_I c\| \geq (L-\zeta)\|c\|$. Finally C_g maps the dense span of the standard basis onto the span of shifts of g. Its invertibility makes the latter dense. □

In Example 6.2, every finite orbit Gram has lower bound $11/20$. For $I = \{-K, \dots, K\}$, the cruder estimate based only on the largest nonzero correlation gives $1 - 2K/5$, which is nonpositive for $K \geq 3$. The certified whole-circle floor retains a positive bound independently of this window size. This improves that particular certificate, not the physical signal space or its approximation error. The full Toeplitz structure is being used instead of discarding it in a largest-entry bound.

Proposition 7.2 (Finite signed spectral cascades)**.** *For finitely many real normalized Littlewood factors p_j and positive integers h_j, suppose $L_j \leq |p_j| \leq U_j$ with $L_j > 0$. The multiplier $Q(t) = \prod_{j=1}^{J} p_j(h_j t)$ has convolution lower and upper bounds $\prod_j L_j$ and $\prod_j U_j$, respectively. No independence of the factors is needed. If p_j has length N_j and $h_1 = 1$, $h_j = \prod_{i<j} N_i$, then Q itself is a normalized real Littlewood polynomial of length $\prod_j N_j$, and $\int |Q|^2 = 1$.*

Proof. The multiplier bounds follow by multiplying the pointwise inequalities and applying the preceding Plancherel argument. In the mixed-radix case, each exponent $0 \leq k < \prod_j N_j$ has a unique representation $k = \sum_j h_j k_j$ with $0 \leq k_j < N_j$. Expanding the product therefore gives one sign, not a sum of colliding coefficients, at every exponent. Its common coefficient modulus is $(\prod_j N_j)^{-1/2}$; Parseval gives normalization. Arbitrary dilations need not have this uniqueness, so their products must not silently be declared normalized Littlewood polynomials. □

These are finite physical spectral-product consequences. They make no claim that the complex multipliers converge along an infinite cascade, or that a product density is the maximal spectral type of a probability transformation. Uniform products and logarithmic criteria belong to the separate conditional layer discussed in the private comparison [6].

7.1 An explicit failure of physical-vector compactness

Proposition 7.3 (Long flat filters can still escape weakly)**.** *Let $N_j \to \infty$, and let $g_j \in \ell^2(\mathbb{Z})$ have exactly N_j nonzero coefficients, each of modulus $N_j^{-1/2}$, at any distinct integer positions. Then $g_j \rightharpoonup 0$, while $\|g_j\| = 1$; no subsequence converges strongly. This remains true after any scalar phase or integer translation. If source-ultraflat consecutive factors are supplied, their orbit Grams can simultaneously have bounds tending to identity for every finite window, without any strong limit of g_j.*

Proof. For finitely supported f, $|\langle g_j, f\rangle| \leq \|f\|_1/\sqrt{N_j} \to 0$, independently of positions and phases. Approximate any $f \in \ell^2$ by a finite vector and use the uniform unit norms to get weak convergence. A strong subsequential limit would have to be the weak limit zero, but would also have norm one. If the factors have two-sided errors tending to zero, Theorem 7.1 gives their Gram bounds; the same weak-convergence argument still applies. □

This is a concrete warning for the physical cyclicity passage, not a new spectral obstruction to the source's independent-length claim. Convergence of modulus or of finite correlations controls no missing phase compactness and supplies no limit unit vector in this embedding.

8 Realization gates, limitations, and further work

The freshly read private comparison [6] distinguishes the following issues, which are retained here. For ordinary rank-one tower factors with positive coefficients, $R(z) = r^{-1/2}\sum_{j=1}^{r} z^{a_j}$, one has $R(1) = \sqrt{r}$ and hence $\||R|^2 - 1\|_\infty \geq r - 1$. Spacer positions do not alter that evaluation. It is not a contradiction to a *signed* Littlewood construction, and the signed factors do not automatically encode ordinary positive tower cuts or probabilities.

A unitary spectral model is not automatically a Koopman operator. Probability normalization, positivity, preservation of multiplication, the constant function, and a compatible tower or other measurable realization must be proved. The natural convolution construction above already operates on an abstract bilateral shift; producing a stable cyclic vector there does not solve Banach's probability-preserving spectral problem.

For moving-window arguments the private comparison requires a common physical Hilbert space, actual unitaries $U_J \to U$ strongly on fixed powers, unit vectors $\phi_J \to \phi$, density of finite approximate cyclic spans, and a transfer budget

$$T_J^* T_J \succeq a_J I, \qquad \|T_J - S_J\|/\sqrt{a_J} \longrightarrow 0,$$

where T_J synthesizes the approximate orbit and S_J the true orbit. Our floor can supply a_J only when the relevant vector's *actual* spectral density is the certified q_J. It supplies none of the other hypotheses. Proposition 7.3 exhibits a natural embedding in which the required vector convergence specifically fails.

Advantages. The join certificate shows which mismatch has a summable spectral cost and which destroys the requested absolute tail altogether. The finite completion theorem separates linear projection error from quadratic energy loss, retains actual sign sparsity, and yields a usable lower-floor inequality. Exact correlations then connect that floor to physical convolution and all finite orbit Grams, without a window-size loss from using only the largest correlation. These deductions use the source's specific continuous-wave and sign-projection mechanism.

Costs and nonclaims. No small numerical residual replaces exact continuity, a certified coefficient bound, or a rigorous quadrature/tail estimate. The constants $M, \Lambda, K, \sigma^{-1}$ can be large for the source's many small intervals; no effective construction threshold or computational speedup is proved. Discrepancy and signed packing remain imported. The zero-offset rounding refinement is classical compactness, the variance bound is classical, and the physical multiplier argument is Plancherel, not a renamed global theorem. No mathematical error in the assigned manuscript is asserted by this scoped read. We retain its common-vector weights, phase coherence, real symmetry, all-length order of choices, and distinction between first boundary cancellation and full-derivative matching.

Attainable next steps. One can certify a concrete wave profile with interval bounds on phase derivatives and exact endpoint continuity, then apply the join budget before spending effort on rounding. The sign deficit can be computed directly instead of using its energy majorant. A supplied sign polynomial can be checked by exact Chebyshev positivity or certified extrema, with the resulting floor passed to a physical convolution implementation. For dynamical applications, the next question is an actual signed realization and moving-vector construction, not another abstract near-identity Gram matrix. These tasks remain separate and may be difficult.

9 Evidence and contribution assessment

The owned read ledger gives literal intervals, unread complements, and pre/post hashes. All 1,333 lines of the nine assigned public text files were read. Fresh private reads comprise selected actual passages in the TR169 comparison (303 lines) and the original TR8, TR248, and TR250 sources (246 lines combined), for 549 private lines in total; other passages and uninspected cited originals remain unread. The companion packing/discrepancy proofs are explicitly imported. Only public titles and URLs were sent to literature search; the private corpus stayed local.

The exact executable checks use rational arithmetic for polynomial and correlation identities, dyadic-rounding bookkeeping, reciprocal join terms, and bounded integer instances. They test implementations of local formulas, not an all-length construction or a Koopman realization. The suite passed 88,177 exact assertions in 11,868 bounded cases; its separate units and domains are recorded in the owned JSON. Parent reviewer checks are not included in author counts.

Applicability stays L3. The independent C3 assessment concerns the explicit join-defect-to-exterior-tail estimate and its finite two-sided completion/physical-floor consequences, together with the precise weak-compactness boundary. The sparse zero-offset refinement and textbook identities alone would be a minor contribution. No C4/C5 claim is made: there is no new globally constructed ultraflat family, quantitative all-length threshold, or Banach-spectrum solution.

References

[1] OpenAI, *Ultraflat real Littlewood polynomials*, October 5, 2026, assigned public cached manuscript. Pinned source commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`; the owned ledger records local hashes and the parent records captured-target authentication.

[2] OpenAI, *Nearly minimal maxima and positive minima of Littlewood polynomials*, October 5, 2026, Lemmas 3.1 and 6.2 as quoted in the assigned source. Their proofs are imported, not independently certified here.

[3] P. Balister, B. Bollobás, R. Morris, J. Sahasrabudhe, and M. Tiba, *Flat Littlewood polynomials exist*, Annals of Mathematics 192 (2020), 977–1004, doi:10.4007/annals.2020.192.3.6. Exact Theorem 1.1 inspected in the publisher text; not a full proof audit.

[4] T. Erdélyi, *On an Erdős Problem about the Maximum Modulus of Littlewood Polynomials on the Unit Circle*, arXiv:2608.00744v1, August 2026. Exact Theorems 2.1–2.2 inspected; not independently re-proved here.

[5] T. Erdélyi, *The sequence of partial sums of a unimodular power series is not ultraflat*, Journal of Approximation Theory 313 (2026), 106219, doi:10.1016/j.jat.2025.106219. Exact Theorem 2.1 inspected in the author's July 20, 2025 text; its Bernstein-factor input remains imported.

[6] M. Subbarao, confidential spectral-problem comparison, `res-math/TR-item-169-astra.tex`, selected block, product, obstruction, and physical moving-window passages freshly read.

[7] M. Subbarao, confidential original spectral analysis, `TR_2026-04-04-0008_V2Ram_Transform_Spectral_AnalysisV2.tex`, lines 172–248 freshly read; Fourier conventions and substitution derivation.

[8] M. Subbarao, confidential original frame manuscript, `TR-rev-1-old-tr248.tex`, lines 142–203 and 398–456 freshly read; physical coordinates, finite scope, and perturbation proof.

[9] M. Subbarao, confidential original finite-moment manuscript, `TR-rev-1-old-tr250.tex`, lines 155–180 and 455–476 freshly read; compatible finite moment inversion and positive Gram construction.